Detection of Abnormalities in the RPE in Long-Chain 3-Hydroxyacyl-CoA Dehydrogenase Deficiency Chorioretinopathy Using Optical Coherence Tomography–Derived Optical Attenuation Coefficients

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Purpose. To detect abnormalities in the RPE in long-chain 3-hydroxyacyl-CoA dehydrogenase deficiency (LCHADD) chorioretinopathy, a novel method using optical attenuation coefficients (OACs) was used to enhance the contrast between regions with high and low attenuation.

Methods. LCHADD patients were scanned over the central macula using two commercial optical coherence tomography (OCT) devices. Fundus autofluorescence (FAF) imaging was performed during the same visit to detect RPE abnormalities based on lipofuscin distribution. OAC was derived from OCT data. En face OAC of the retina (internal limiting membrane to Bruch's membrane [BM]), en face OCT of the RPE (16 µm above the BM to the BM), and corresponding FAF images were compared to evaluate the ability of the OAC to detect RPE abnormalities. Pairwise comparisons were performed across all LCHADD stages, along with correlations with traditional clinical metrics including LCHADD severity, best-corrected visual acuity, and plasma acylcarnitines.

Results. Forty-three eyes from 22 LCHADD patients were evaluated and categorized into 6 stages. OAC images of the retina revealed RPE defects consistent with FAF, with clearer and more detailed visualization of RPE disruption. Compared with OCT images, OAC images facilitated visualization of pathological changes and provided a better visual comparison between the preserved and degenerated RPE at advanced stages. The mean retinal OAC decreased significantly ($P < 0.001$) with disease progression and differed significantly across disease stages (all $P \leq 0.002$), showing strong correlations with traditional clinical metrics.

Conclusions. OCT-based OAC is a promising tool for both the quantitative and qualitative assessment of RPE abnormalities in LCHADD chorioretinopathy.

Keywords: OCT, OAC, LCHADD, RPE

Long-chain 3-hydroxyacyl-CoA dehydrogenase deficiency (LCHADD) is a rare fatty acid oxidation disorder associated with progressive chorioretinopathy that ultimately leads to vision loss.[1,2] Although the pathophysiology of LCHADD-associated chorioretinopathy remains incompletely understood, many studies suggest that RPE dysfunction occurs early in the disease process.[3,4] The RPE provides metabolic support to photoreceptors, and phagocytoses shed photoreceptor outer segments, thereby enabling the clearance and recycling of cellular waste products.[5,6] In addition, the RPE participates in the vision cycle and regenerates the retinoids required for phototransduction.[7] RPE melanin granules further protect the retina by absorbing excess light and scavenging light-induced free radicals.[8] The disruption of RPE function can compromise photoreceptor survival and retinal homeostasis, leading to progressive photoreceptor and retinal degeneration and vision impairment in individuals with LCHADD.[9,10]

Optical coherence tomography (OCT) and fundus autofluorescence (FAF) are standard, noninvasive imaging modalities, widely and complementarily used in clinical diagnostics and scientific research, including studies of LCHADD. OCT offers reliable, depth-resolved retinal structural information,[11,12] including detailed visualization of the RPE layer. However, structural OCT has limitations in detecting the early stages of LCHADD,[13,14] because RPE cells may become dysfunctional before cell death or tissue thinning occurs.[15] FAF, meanwhile, provides a lipofuscin density map

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within RPE cells, enabling the visualization of RPE abnormalities.[16–18] Despite its widespread clinical use and qualitative value, FAF remains limited in its ability to reliably quantify and compare autofluorescence intensities across individuals or over time. This is primarily due to variability in imaging conditions and the absence of standardized calibration protocols.[19,20]

The optical attenuation coefficient (OAC) is a material property that quantifies opacity, enabling the detection of subtle tissue alterations that may precede overt changes, such as cell death. OAC measurements can be derived directly from OCT scans by quantifying the light intensity as it propagates through tissue.[21–25] This approach enhances the contrast between regions of high and low scattering and can be used to localize and quantify pigment-related optical attenuation in the macula.[26] Because the RPE contains a relatively high pigment concentration, it represents a suitable target for OAC analysis. OAC analysis has consequently been applied to quantitatively assess and distinguish abnormal from healthy RPE in AMD,[27] measure RPE thickness in AMD,[26,28] and localize hyperpigmentation along the RPE in glaucoma.[29] RPE cells contain two major pigment classes: melanin and lipofuscin. OAC values are strongly influenced by melanin-associated optical attenuation, because melanin is a major light absorber and a key contributor to attenuation in the RPE.[30,31] Consequently, OAC measurements provide information on RPE attenuation and may serve as a surrogate marker of RPE integrity in LCHADD chorioretinopathy by reflecting melanin-related optical attenuation.

In this study, we evaluated the utility of OAC measurements for detecting RPE abnormalities in LCHADD chorioretinopathy. En face OAC of the retina was compared with FAF and structural OCT to assess their ability to capture RPE alterations linked to melanin-related optical attenuation. We further performed quantitative analyses of the mean retinal OAC across LCHADD severity stages and examined its correlations with LCHADD severity, best-corrected visual acuity (BCVA), and plasma acylcarnitines (a traditional LCHADD biomarker of partially oxidized long-chain fatty acids).

Methods

Study Inclusion Criteria and Examination Procedure

Participants were recruited for the Natural History of LCHADD Retinopathy study, a two-site prospective cohort study conducted at the Oregon Health & Science University (OHSU) and the University of Pittsburgh Medical Center (UPMC). The study was approved by the Institutional Review Board of the UPMC (IRB# PRO19040142). The OHSU deferred oversight to the UPMC. The participants included in this report were all evaluated at the Casey Eye Institute at OHSU. Informed consent was obtained from all participants, and the study adhered to the Declaration of Helsinki.

LCHADD was confirmed by a review of medical records and the identification of two pathogenic mutations in the HADHA gene. These patients underwent a comprehensive eye examination, including measurements of BCVA, spherical equivalent, and axial length (IOL Master 500; Carl Zeiss Meditec, Dublin, CA, USA), as well as full-field ERG, FAF (Optos 200Tx; Optos PLC, Marlborough, MA, USA), and spectral-domain OCT imaging performed using two systems (Avanti and Solix; Visionix USA, Lombard, IL, USA). The spherical equivalent was calculated as a spherical power plus one-half of the cylindrical power. Axial length was incorporated into the image analysis to correct for ocular magnification effects. Fundus images and clinical data were reviewed by an ophthalmologist, and eyes were classified into six stages of LCHADD chorioretinopathy according to the updated staging system proposed by Wongchaisuwat et al.[13] This classification scheme includes the following: stage 1, both the retina and choroid appear normal; stage 2A, normal to thickened RPE–Bruch's membrane (BM) complex in the macula, with mild attenuation of the ellipsoid zone/RPE and a normal choroid; stage 2B, similar changes but changes can extend beyond the vascular arcades into the periphery; stage 3A, progressive outer retinal attenuation with a normal or mildly thinned choroid; stage 3B, more extensive progressive atrophy of both the outer retina and choroid but sparing the central fovea; and stage 4 extensive outer retinal and choroidal atrophy, with the central fovea largely spared.

The patients were excluded when the following appeared: (1) other ocular or systemic conditions affecting the retina rather than LCHADD, (2) previous intraocular surgery except for uncomplicated cataract extraction with posterior chamber intraocular lens implantation, (3) the inability to maintain fixation during the OCT procedure, and (4) OCT scans with a low signal strength index (<55), significant motion artifacts, defocusing, or off-center positioning.

OCT Imaging

Foveal avascular zone–centered scans were collected from patients with LCHADD chorioretinopathy using two commercial spectral-domain OCT systems: (1) Avanti with an 840-nm central wavelength and 70-kHz axial scan rate, covering a 6 × 6-mm area, and (2) Solix with an 840-nm central wavelength and 120-kHz axial scan rate, covering a 6.4 × 6.4-mm area. On the Avanti system, each B-scan is composed of 400 A-lines, and each data volume consists of 400 B-frames. On the Solix system, each B-scan is composed of 512 A-lines, and each data volume consists of 512 B-frames.

The OCT volumes were processed using the Center for Ophthalmic Optics & Lasers-Angiography Reading Toolkit image processing software.[32] Retinal layer boundaries, including the vitreous and the internal limiting membrane (ILM), as well as the BM, were automatically segmented from the structural OCT volume. Any obvious segmentation errors that occurred were corrected manually by an experienced OCT grader. All layer segmentations were subsequently reviewed by an ophthalmologist with experience in OCT imaging, and additional corrections were made as needed. However, some pixel segmentation errors cannot be completely avoided, as they may not be reliably identified by human observers (see Discussion). En face OCT images of the RPE were generated using a mean projection from a slab extending from 16 µm above the BM to the BM. All OCT images were adjusted to correct for the effect of axial length on linear magnification.[33] The region of interest was defined within a 6 × 6-mm square area centered on the foveal avascular zone.

OAC Calculation

The OAC values were derived from OCT data, and several methods have been used to calculate the OAC.[9,24,25] In this study, we used the depth-resolved method described by

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Vermeer et al.[23] to convert OCT data into OAC volumes:

$$\mu(z) = \frac{I(z)}{2\Delta \sum_{z+1}^{\infty} I(z)}, \quad (1)$$

where μ(*z*) denotes the OAC at the *z*th pixel along the depth direction, *I*(*z*) represents the OCT signal intensity (linear scale), and Δ refers to the axial size of a pixel (approximately 3 μm in this study). Given the assumption of full attenuation within the imaging range, $\sum_{z+1}^{\infty} I(z)$ can be calculated by summing the OCT intensities of all pixels below the *z*th pixel. A manufacturer-provided equation was used to convert logarithmic OCT data back to a linear scale.

The segmentation boundaries of the ILM and BM derived from OCT data were applied to the corresponding volumetric OAC data. En face OAC of the retina was generated using a mean projection within the slab from the ILM to the BM. All OAC images were also adjusted to correct for the effect of axial length on linear magnification. The mean retinal OAC was calculated by averaging the OAC values within the region of interest on the OAC images.

Quantitative Evaluation of RPE Detection: OCT Reflectance vs. OAC

To assess how well OCT reflectance and the OAC identify RPE defects, we evaluated the segmentation of the preserved RPE in the en face images from each modality by comparing it with a ground truth manually delineated across all B-scans. Only eyes with advanced-stage LCHADD (stages 3–4) (Supplementary Fig. S1) were used for this evaluation, because the RPE regions could be distinctly identified and the ground truth could be confidently generated in these cases. Spatial overlap between each binary mask and the ground truth was quantified using intersection over union.

Plasma Acylcarnitines Acquisition and Processing

Elevated plasma long-chain 3-hydroxyacylcarnitines serve as screening and diagnostic biomarkers for LCHADD, and we have reported a relationship between elevated acylcarnitines and lower visual acuity.[34] Plasma from a fasting blood sample was analyzed for acylcarnitines by liquid chromatography tandem mass spectrometry, and the LCHADD-specific species were quantified for all participants.[35,36] A principal components analysis (PCA) was used to summarize these acylcarnitine species and reduce data dimensionality,[37] representing overall plasma long-chain 3-OH acylcarnitines concentrations for each participant. The first principal component (PCA1) explained 88% of the variation in acylcarnitines and was used as a biomarker of disease severity, with higher PCA1 values indicating higher levels of long-chain 3-hydroxyacylcarnitines and lower LCHADD enzymatic activity.

Statistical Methods

The statistical analyses were performed with SPSS 29.0 (IBM Corp., Armonk, NY, USA). The Shapiro–Wilk test was used to assess the normality of the data distribution. The Mann–Whitney *U* test was used to compare two groups, and the Bonferroni correction was applied to adjust for multiple pairwise comparisons. Spearman's rank-order correlation analysis was used to assess the correlations between the mean retinal OAC and the mean RPE reflectance and LCHADD severity, BCVA, and plasma acylcarnitines. Because both eyes from many participants were included in the analysis, linear mixed models with random intercepts for participants were used to account for the intrasubject correlation between eyes.

Results

Study Population

Of the 40 participants diagnosed with LCHADD, 22 were included in this analysis, and 43 eyes were evaluated in this study. Participants ranged in age from 2 to 30 years at the time of evaluation. One eye was excluded from the analysis due to poor scan quality (signal strength index of <55). Based on the LCHADD scoring system, which incorporated a comprehensive assessment including fundus changes, OCT characteristics, FAF, and full-field ERG findings,[11] the included eyes were classified into one of six stages of LCHADD chorioretinopathy (Table 1).

RPE Visualization Comparison

Representative cross-sectional OCT images and corresponding cross-sectional OAC images from eyes with LCHADD at different stages were investigated (Fig. 1). The RPE regions exhibited more distinct attenuation patterns on cross-sectional OAC images than on conventional OCT reflectance images across disease stages, likely due to differences in the optical attenuation properties of melanin-containing RPE cells relative to the surrounding retinal tissues. The OAC signal highlighted melanin-related RPE alterations, particularly in regions where the RPE structural architecture is preserved (*blue arrows*) but melanin is compromised (*green arrows*), thereby improving the visualization of RPE viability.

En face OAC images of the retina were compared with FAF and en face OCT images of the RPE for visualizing RPE defects (Fig. 2). FAF imaging revealed lipofuscin distribution and detected RPE defects across different stages of LCHADD. However, its signal was weak in the 6 × 6-mm macular area, limiting its effectiveness in this region.

En face OAC images of the retina (Fig. 2, row 2) closely corresponded with FAF images in identifying RPE defects across different disease severities. For instance, in stage 1, OAC images revealed central abnormalities, consistent with FAF findings. In stage 2B, both OAC and FAF images demonstrated a speckled hypoautofluorescence pattern. In stage 3A, patchy RPE atrophy was observed in both imaging modalities. In stage 4, extensive RPE loss was detected, with only a residual central island visible in both OAC and FAF images. Thus, the OAC measurements reflected RPE alterations comparable with those observed with FAF. In OAC images, bright areas represented regions with relatively preserved melanin-related attenuation, suggesting preserved RPE viability, whereas dark areas indicated reduced attenuation, which corresponds with RPE degeneration or a loss of melanin content. Therefore, these OAC images demonstrated RPE alteration patterns consistent with the FAF findings.

En face OCT images of the RPE revealed structural changes in RPE cells (Fig. 2, row 3). RPE reflectance appeared normal at stage 1, with only minimal changes detectable in stages 2A and 2B, indicating a limited ability to identify early-stage RPE abnormalities in LCHADD. In advanced stages, particularly at stage 4, the en face OCT

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Table 1. Participant Characteristics

Parameter	LCHADD					
	Stage 1	Stage 2A	Stage 2B	Stage 3A	Stage 3B	Stage 4
Participants	6	6	4	1	2	3
Eyes, no.	12	12	7	2	4	6
Age, years	12.1 ± 3.5	10.9 ± 4.9	14.6 ± 3	18.8	26.4 ± 1.9	22.4 ± 6.3
SSI	82.67 ± 7.00	68.58 ± 8.06	69.57 ± 4.69	74.00 ± 2.83	70.00 ± 8.04	80.50 ± 6.77
SE, D	0.16 ± 0.24	−0.13 ± 0.87	−1.16 ± 1.39	−1.06 ± 0.09	−3.81 ± 0.52	−6.50 ± 2.79
AL, mm	23.22 ± 0.26	22.62 ± 0.81	24.29 ± 0.48	23.42 ± 0.13	23.14 ± 0.46	25.45 ± 0.74
BCVA, logMAR	−0.11 ± 0.07	−0.04 ± 0.13	0.08 ± 0.11	0.20	0.87 ± 0.84	0.77 ± 0.48
Plasma acylcarnitines PCA1	−1.89 ± 2.04	−1.17 ± 1.24	2.39 ± 0.46	2.37	2.27 ± 1.25	2.00 ± 0.64

AL, axial length; D, diopters ; SE, spherical equivalent; SSI, signal strength index.

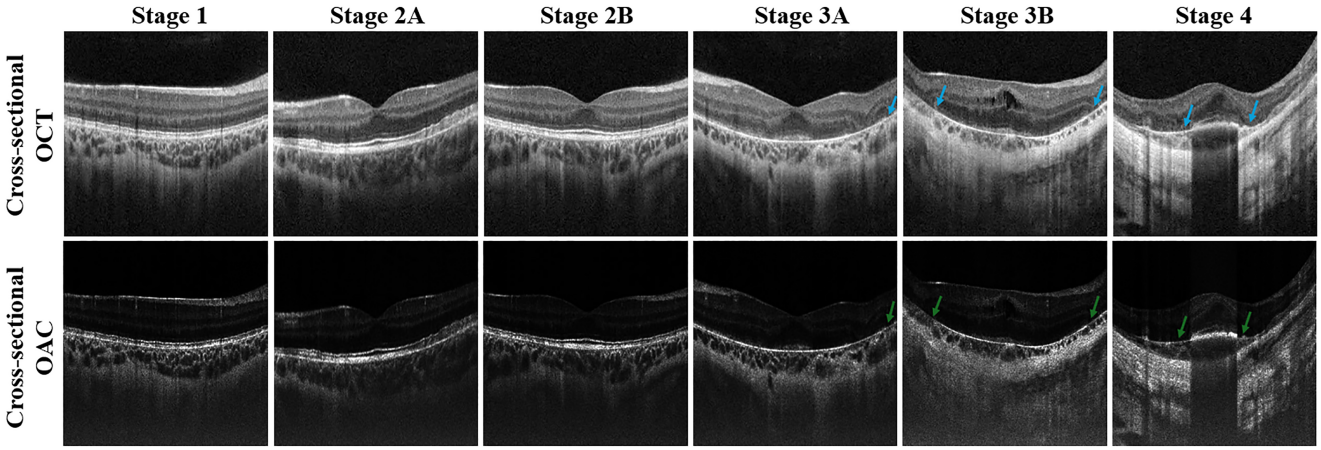


Figure 1. Representative cross-sectional OCT images (*row 1*) and corresponding cross-sectional OAC images (*row 2*) from eyes with LCHADD at different stages. Compared with conventional OCT reflectance images, cross-sectional OAC images provide attenuation-based information that highlights RPE alterations. Therefore, RPE signal alterations were more readily identified in OAC images (*green arrows*) than in OCT images (*blue arrows*) across different LCHADD stages.

images of the RPE became blurred, hindering assessment of the remaining tissue structure. The images also exhibited low contrast at stages 3A and 3B, complicating the distinction between the areas of preserved or lost RPE. These issues likely resulted from attenuation or loss of the RPE layer (Fig. 1, stages 3A–4), causing segmentation errors and interference from adjacent layers. As a result, en face OCT images of the RPE provided limited visualization of RPE defects.

Overall, retinal en face OAC images showed qualitative spatial correspondence with FAF findings while offering enhanced visualization of differences between preserved and degenerated RPE regions compared with conventional en face OCT images, particularly in advanced disease stages. Additionally, en face OAC images of the retina identified RPE abnormalities as early as stage 1.

Quantitative RPE Defect Assessment

OAC images demonstrated significantly greater spatial overlap with the preserved RPE mask based on OCT B-scans than conventional en face OCT images (intersection over union: 0.61 ± 0.19 vs. 0.22 ± 0.21; $P < 0.001$), indicating that OAC images more closely represented the spatial distribution of the preserved RPE.

In addition to assessing the preserved RPE regions, global image characteristics were quantified by measuring the mean retinal OAC and mean RPE reflectance across different LCHADD stages. Within the 6 × 6-mm macular area, no differences were observed between stages 2A and 2B or between stages 3A and 3B on OCT images. This result was not unexpected; the updated staging system proposed by Wongchaisuwat et al.[13] distinguishes these substages using multimodal findings that extend beyond macula-centered OCT imaging, including wide-field FAF, choroidal findings on OCT, and full-field ERG. Therefore, stages 2A and 2B were combined as stage 2, and stages 3A and 3B were combined as stage 3 for the statistical analysis.

The mean retinal OAC values progressively decreased with increasing LCHADD severity (Table 2). Comparisons between stages using the Mann–Whitney *U* test indicated significant differences across stages (Fig. 3), particularly between early stages (stage 1 vs. stage 2). In contrast, the mean RPE reflectance increased nonsignificantly from stage 1 to stage 2, decreased in stages 3 and 4, and showed no significant difference between stage 3 and stage 4. Additionally, the mean retinal OAC exhibited stronger correlations with LCHADD severity, BCVA, and plasma acylcarnitines than the mean RPE reflectance (Table 3). The mean retinal OAC performed better in quantitatively differentiating LCHADD severity stages and was highly correlated with traditional clinical metrics, suggesting its potential as a more sensitive biomarker for monitoring disease progression. After adjust-

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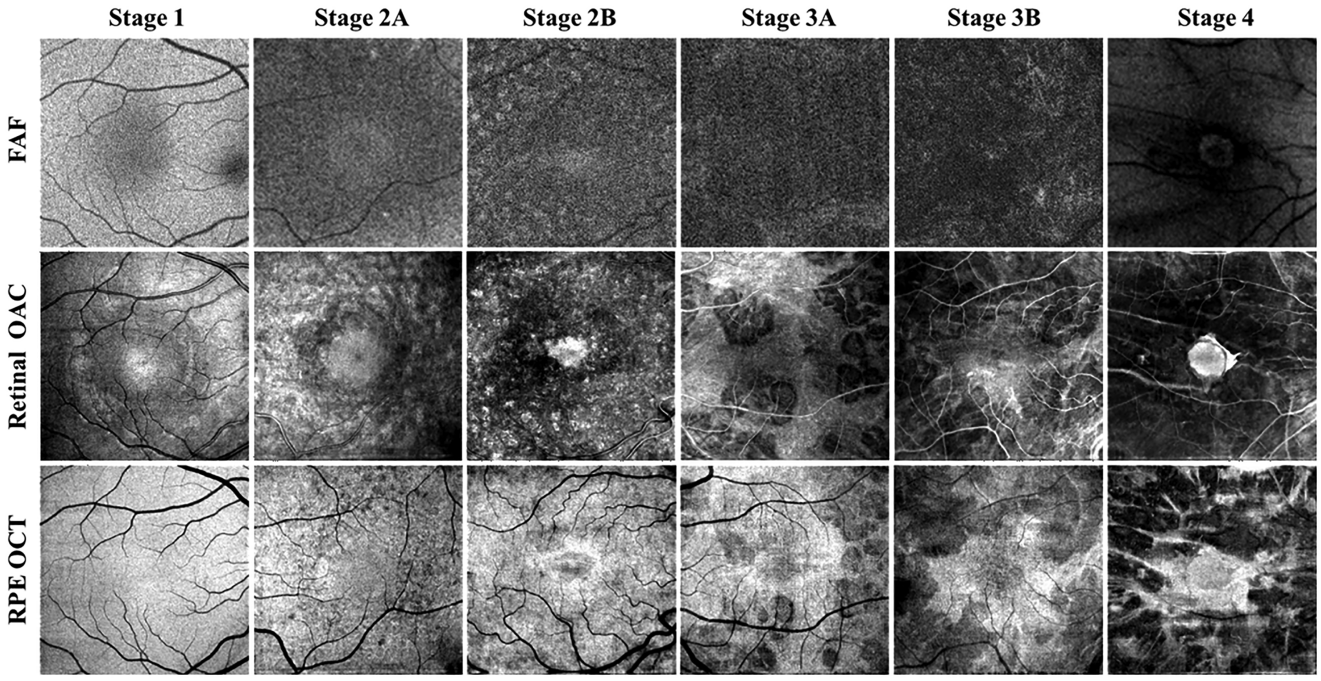


Figure 2. Multimodal comparison of RPE defect visualization across different severity stages of LCHADD chorioretinopathy. FAF images (*row 1*) reveal RPE abnormalities; however, the signal is weak in the 6 × 6-mm macular area, limiting its effectiveness. En face OAC of the retina (*row 2*) corresponds with FAF images and provides enhanced visualization compared with en face OCT of the RPE (*row 3*), offering more intuitive visualization of RPE integrity in early stages. En face OCT of the RPE reveals structural details but fails to detect early-stage RPE defects and does not clearly delineate their boundaries in advanced stages.

Table 2. Mean Retinal OAC and Mean RPE Reflectance Across LCHADD Severity Stages

Parameters	Stage 1	Stage 2	Stage 3	Stage 4
Mean retinal OAC, mm^{-1}	6.17 ± 0.65	4.76 ± 0.47	2.41 ± 0.75	1.44 ± 0.21
Mean RPE reflectance	0.65 ± 0.02	0.67 ± 0.03	0.55 ± 0.04	0.49 ± 0.03

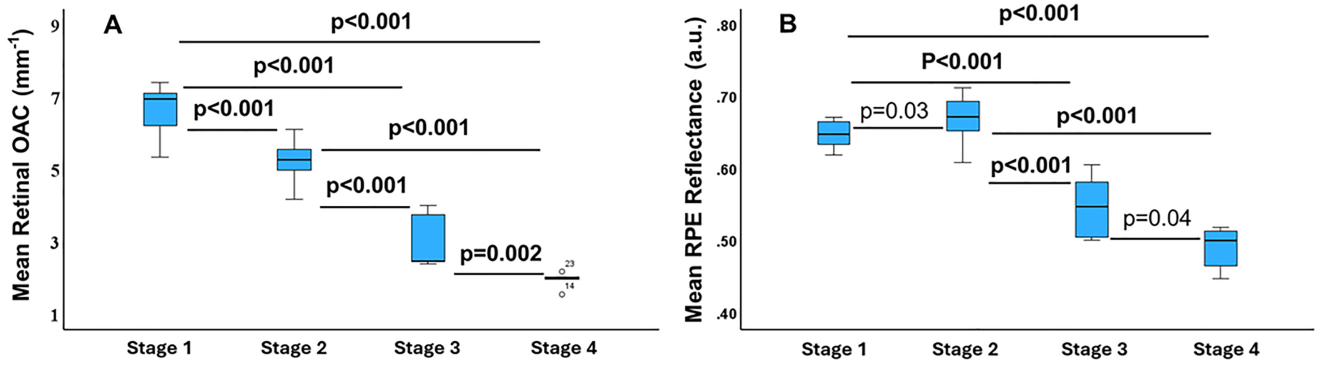


Figure 3. Boxplot of the mean retinal OAC (**A**) and mean RPE reflectance (**B**) for four stages of LCHADD. Statistical differences between groups were analyzed using the Mann–Whitney *U* test with Bonferroni correction. A *P* value of <0.008 (0.05/6.00, accounting for six pairwise comparisons among four groups) was considered statistically significant and is indicated in bold.

ing for intrasubject correlation using a linear mixed model, the mean retinal OAC was significantly associated with LCHADD severity, BCVA, and plasma acylcarnitines (all $P < 0.001$), whereas the mean RPE reflectance was significantly associated with LCHADD severity and BCVA ($P < 0.001$ for both associations) but not with plasma acylcarnitines.

Discussion

In this study, we assessed RPE abnormalities in patients with LCHADD chorioretinopathy, evaluating both qualitative and quantitative pathological changes by measuring the melanin-related pigment optical attenuation derived from OCT data. Our results demonstrated that (1) en face OAC

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Table 3. Spearman Correlation Coefficients of OAC and RPE OCT Parameters With LCHADD Severity, BCVA, and Plasma Acylcarnitines

	Mean Retinal OAC	Mean RPE Reflectance
LCHADD severity	−0.89**	−0.52**
BCVA	0.78**	0.42*
Plasma acylcarnitines PCA1	−0.64**	−0.37

With Bonferroni correction, a P value of <0.008 was considered significant.
* $P < 0.005$.
** $P < 0.001$.

images of the retina provided attenuation-based information regarding RPE alterations in LCHADD within the 6 × 6-mm macular area, with findings consistent with FAF images, (2) OAC images revealed RPE alterations in early-stage LCHADD and demonstrated improved delineation of RPE alterations compared with structural OCT in advanced stages, (3) the mean retinal OAC demonstrated better quantitative differentiation of the LCHADD stages compared with the mean RPE reflectance, and (4) the mean retinal OAC demonstrated stronger correlations with LCHADD severity, BCVA, and plasma acylcarnitines than mean RPE reflectance.

We demonstrated that en face OAC imaging provides useful information regarding RPE integrity and enables the visualization of RPE alterations. Within the 6 × 6-mm macular area, en face OAC images demonstrated RPE abnormalities that showed spatial correspondence with those observed on FAF. These findings indicate that OAC images can capture RPE alterations (Fig. 2, rows 1 and 2). Additionally, the FAF signal intensity within the central macular region was insufficient to identify preserved RPE regions in many cases; therefore, OAC images provided supplementary information on RPE alterations. These findings highlight the potential of en face OAC imaging as a surrogate marker of RPE integrity by assessing melanin-associated optical attenuation.

OAC measurements provide enhanced visualization of RPE pathological alterations and improved delineation of preserved RPE regions in LCHADD chorioretinopathy compared with conventional structural OCT. En face OAC images of the retina can reveal RPE changes as early as stage 1 by detecting melanin-associated changes in the RPE cells (Fig. 2, rows 2 and 3). In the regions where RPE cells have lost pigment due to degeneration or cell death, the OAC values decrease markedly, producing darker regions on the OAC images.[26] In contrast, areas with preserved or partially preserved RPE show higher OAC values and appear brighter, enabling differentiation between the degenerated and the preserved RPE. Although numerous studies have reported structural RPE atrophy by detecting RPE thinning on cross-sectional OCT images at severe stages,[38–40] commercial structural OCT may have limited sensitivity for detecting subtle RPE abnormalities at earlier stages. Previous histopathology and cell culture studies have suggested that long-chain fatty acid oxidation affects RPE health and function early in LCHADD chorioretinopathy.[4,41–43] These OAC findings provide in vivo evidence supporting these earlier reports by detecting RPE changes before structural thinning becomes apparent. Additionally, en face OAC imaging facilitated the identification of preserved RPE regions by enhancing the visualization of the differences between the preserved and the degenerated RPE, particularly in advanced disease stages.

En face OAC generated from a whole-retinal slab extending from the ILM to the BM may be less sensitive to pixel-level segmentation errors than en face OCT images generated from a thin RPE slab. By measuring optical attenuation across the whole retinal slab, OAC images generated from this slab may reduce the impact of small segmentation inaccuracies in the resulting en face images. In contrast, similar pixel-level segmentation errors can have a relatively greater impact on a thin RPE slab because the slab is only a few pixels thick. The accurate delineation of the RPE layer is particularly challenging in the advanced stages of LCHADD chorioretinopathy, where marked RPE thinning or loss is present. Typically, the upper boundary of the RPE slab is defined at a fixed distance above the BM based on anatomical measurements of the normal human retina.[44,45] However, in advanced stages with RPE thinning, this approach may incorporate signals from adjacent retinal structures, thereby reducing visual clarity. In severe stages, BM boundaries become indistinct, allowing deeper choroidal signals to interfere with RPE visualization. These factors may contribute to blurred or less distinct en face OCT images. Therefore, whole-retinal slab en face OAC imaging may provide a more robust approach for visualizing RPE abnormalities in advanced LCHADD chorioretinopathy, particularly when the accurate delineation of a thin RPE slab is challenging.

We demonstrated that the mean retinal OAC measurements provide a superior quantitative performance in distinguishing the stages of LCHADD chorioretinopathy. Most previous studies on LCHADD chorioretinopathy have focused on qualitative morphological descriptions and pathological observations.[46,47] To our knowledge, no effective quantitative approach has been established to differentiate LCHADD severity using noninvasive imaging. In contrast, our results revealed a progressive decrease in the mean retinal OAC values as the LCHADD severity increased (Table 2), with significant interstage differences, particularly at earlier stages (Figure 3), where differentiation using the mean reflectance measurements from the RPE slab remains difficult. Additionally, the mean retinal OAC correlated strongly with LCHADD severity, BCVA, and plasma acylcarnitines (Table 3).

Although the OAC approach is evident, this study has a few limitations. (1) Because LCHADD chorioretinopathy is a rare disease, the sample size was small, with most cases in the early stages and fewer in advanced stages, resulting in an imbalanced dataset. (2) Participants' ages were not matched, making it difficult to exclude potential age-related effects on LCHADD progression. (3) Because the field of view was limited, differences between stages 3A and 3B could not be accurately demonstrated. (4) Although several image-quality control and correction procedures were implemented, residual effects related to individual ocular characteristics, such as refractive error and ocular media status, cannot be completely excluded and may have influenced OCT-derived quantitative measurements. (5) Two OCT systems were used for data acquisition during the study period. Although both systems were manufactured by the same company and are based on spectral-domain OCT technology, potential interdevice measurement variability cannot be entirely ruled out. (6) Disease severity assessment and image interpretation were performed by a senior retinal specialist, which may introduce observer bias and prevents

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the assessment of intergrader reproducibility. These limitations should be addressed in future studies through larger cohorts, longitudinal follow-up, standardized imaging protocols, and evaluation of intergrader agreement to further validate the reproducibility and clinical applicability of OAC-based assessment.

Conclusions

Our findings demonstrated that the OAC method enabled both qualitative and quantitative assessment of RPE abnormalities in LCHADD chorioretinopathy. En face OAC imaging of the retina revealed RPE defects that correspond with FAF images while enhancing visualization of RPE alterations compared with en face OCT imaging of the RPE. The mean retinal OAC is sensitive to LCHADD disease progression, showing a significant decrease with increasing disease severity. Furthermore, the mean retinal OAC exhibits strong correlations with traditional clinical metrics, including LCHADD severity, BCVA, and plasma acylcarnitines. Collectively, these results suggest that the OAC approach may serve as a sensitive imaging biomarker for detecting RPE abnormalities and monitoring disease progression in LCHADD chorioretinopathy.

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